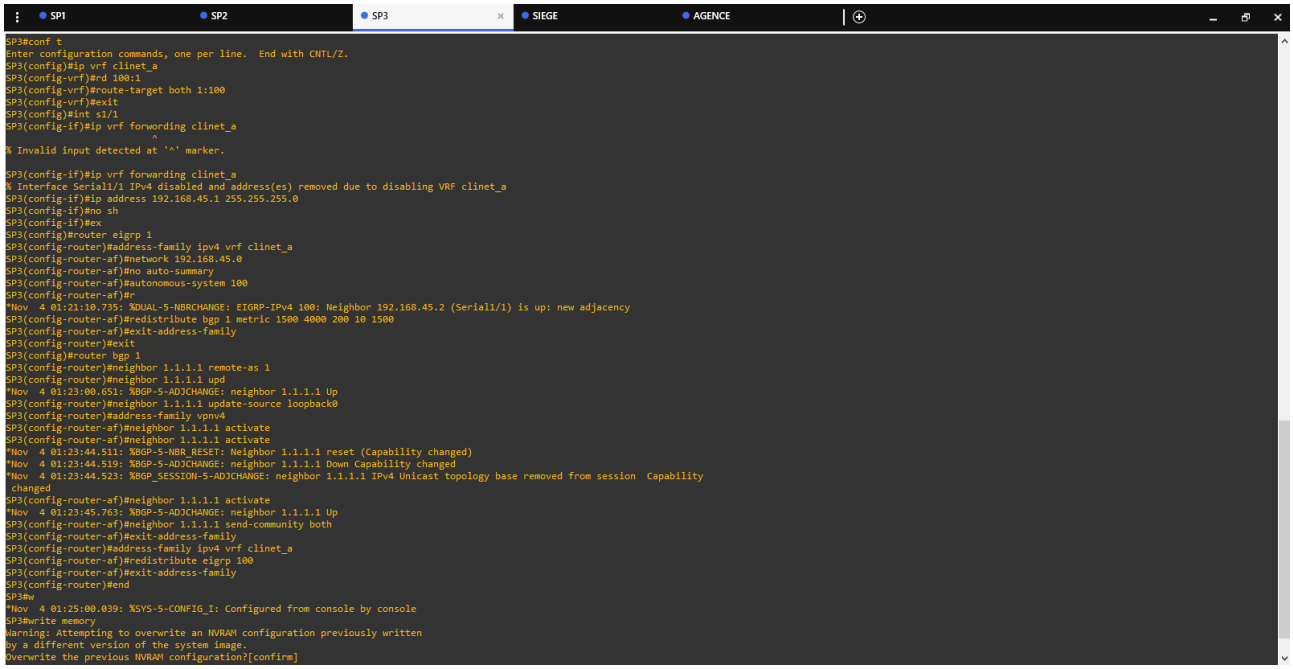
TP2

Captures








```
SP3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SP3(config)#ip vrf clinet_a
SP3(config-vrf)#rd 100:1
SP3(config-vrf)#route-target both 1:100
SP3(config-vrf)#exit
SP3(config)#int s1/1
SP3(config-if)#ip vrf forwording clinet_a
                     ^
% Invalid input detected at '^' marker.

SP3(config-if)#ip vrf forwarding clinet_a
% Interface Serial1/1 IPv4 disabled and address(es) removed due to disabling VRF clinet_a
SP3(config-if)#ip address 192.168.45.1 255.255.255.0
SP3(config-if)#no sh
SP3(config-if)#ex
SP3(config)#router eigrp 1
SP3(config-router)#address-family ipv4 vrf clinet_a
SP3(config-router-af)#network 192.168.45.0
SP3(config-router-af)#no auto-summary
SP3(config-router-af)#autonomous-system 100
SP3(config-router-af)#r
*Nov  4 01:21:10.735: %DUAL-5-NBRCHANGE: EIGRP-IPv4 100: Neighbor 192.168.45.2 (Serial1/1) is up: new adjacency
SP3(config-router-af)#redistribute bgp 1 metric 1500 4000 200 10 1500
SP3(config-router-af)#exit-address-family
SP3(config-router)#exit
SP3(config)#router bgp 1
SP3(config-router)#neighbor 1.1.1.1 remote-as 1
SP3(config-router)#neighbor 1.1.1.1 upd
*Nov  4 01:23:00.651: %BGP-5-ADJCHANGE: neighbor 1.1.1.1 Up
SP3(config-router)#neighbor 1.1.1.1 update-source loopback0
SP3(config-router)#address-family vpnv4
SP3(config-router-af)#neighbor 1.1.1.1 activate
SP3(config-router-af)#neighbor 1.1.1.1 activate
*Nov  4 01:23:44.511: %BGP-5-NBR_RESET: Neighbor 1.1.1.1 reset (Capability changed)
*Nov  4 01:23:44.519: %BGP-5-ADJCHANGE: neighbor 1.1.1.1 Down Capability changed
*Nov  4 01:23:44.523: %BGP_SESSION-5-ADJCHANGE: neighbor 1.1.1.1 IPv4 Unicast topology base removed from session  Capability
 changed
SP3(config-router-af)#neighbor 1.1.1.1 activate
*Nov  4 01:23:45.763: %BGP-5-ADJCHANGE: neighbor 1.1.1.1 Up
SP3(config-router-af)#neighbor 1.1.1.1 send-community both
SP3(config-router-af)#exit-address-family
SP3(config-router)#address-family ipv4 vrf clinet_a
SP3(config-router-af)#redistribute eigrp 100
SP3(config-router-af)#exit-address-family
SP3(config-router)#end
SP3#w
*Nov  4 01:25:00.039: %SYS-5-CONFIG_I: Configured from console by console
SP3#write memory
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
```



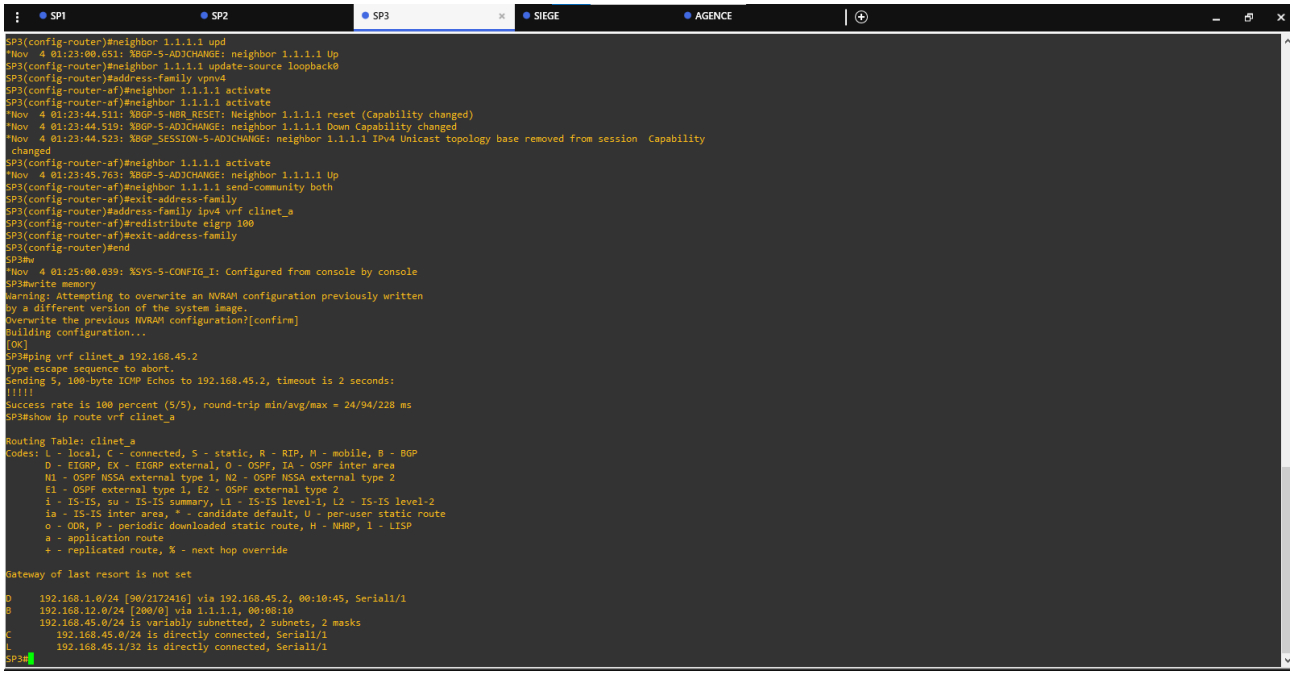

```
SP1#
SP1#
*Nov  4 01:23:00.475: %BGP_SESSION-5-ADJCHANGE: neighbor 3.3.3.3 VPNv4 Unicast topology base removed from session  Capabilit
y changed
*Nov  4 01:23:00.479: %BGP-5-ADJCHANGE: neighbor 3.3.3.3 Up
SP1#
*Nov  4 01:23:44.667: %BGP-5-NBR_RESET: Neighbor 3.3.3.3 reset (Peer closed the session)
*Nov  4 01:23:44.667: %BGP-5-ADJCHANGE: neighbor 3.3.3.3 Down Peer closed the session
*Nov  4 01:23:44.671: %BGP_SESSION-5-ADJCHANGE: neighbor 3.3.3.3 IPv4 Unicast topology base removed from session  Peer close
d the session
SP1#
*Nov  4 01:23:45.799: %BGP-5-ADJCHANGE: neighbor 3.3.3.3 Up
SP1#ping vrf clinet_a 192.168.12.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 64/128/288 ms
SP1#show ip route vrf clinet_a

Routing Table: clinet_a
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

B     192.168.1.0/24 [200/2172416] via 3.3.3.3, 00:03:08
      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.12.0/24 is directly connected, Serial1/1
L        192.168.12.1/32 is directly connected, Serial1/1
B     192.168.45.0/24 [200/0] via 3.3.3.3, 00:03:08
SP1#show bgp vpnv4 unicast all summary
BGP router identifier 1.1.1.1, local AS number 1
BGP table version is 6, main routing table version 6
3 network entries using 468 bytes of memory
3 path entries using 240 bytes of memory
3/3 BGP path/bestpath attribute entries using 456 bytes of memory
3 BGP extended community entries using 750 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1914 total bytes of memory
BGP activity 3/0 prefixes, 3/0 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
3.3.3.3         4            1      11      10        6    0    0 00:04:38        2
SP1#
```


SP1
SP2
SP3
SIEGE
AGENCE
SP3(config-router)#neighbor 1.1.1.1 upd
*Nov 4 01:23:00.651: %BGP-5-ADJCHANGE: neighbor 1.1.1.1 Up
SP3(config-router)#neighbor 1.1.1.1 update-source loopback0
SP3(config-router)#address-family vpnv4
SP3(config-router-af)#neighbor 1.1.1.1 activate
SP3(config-router-af)#neighbor 1.1.1.1 activate
*Nov 4 01:23:44.511: %BGP-5-NBR_RESET: Neighbor 1.1.1.1 reset (Capability changed)
*Nov 4 01:23:44.519: %BGP-5-ADJCHANGE: neighbor 1.1.1.1 Down Capability changed
*Nov 4 01:23:44.523: %BGP_SESSION-5-ADJCHANGE: neighbor 1.1.1.1 IPv4 Unicast topology base removed from session Capability
changed
SP3(config-router-af)#neighbor 1.1.1.1 activate
*Nov 4 01:23:45.763: %BGP-5-ADJCHANGE: neighbor 1.1.1.1 Up
SP3(config-router-af)#neighbor 1.1.1.1 send-community both
SP3(config-router-af)#exit-address-family
SP3(config-router)#address-family ipv4 vrf clinet_a
SP3(config-router-af)#redistribute eigrp 100
SP3(config-router-af)#exit-address-family
SP3(config-router)#end
SP3#w
*Nov 4 01:25:00.039: %SYS-5-CONFIG_I: Configured from console by console
SP3#write memory
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
SP3#ping vrf clinet_a 192.168.45.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/94/228 ms
SP3#show ip route vrf clinet_a
Routing Table: clinet_a
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override
Gateway of last resort is not set
D 192.168.1.0/24 [90/2172416] via 192.168.45.2, 00:10:45, Serial1/1
B 192.168.12.0/24 [200/0] via 1.1.1.1, 00:08:10
192.168.45.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.45.0/24 is directly connected, Serial1/1
L 192.168.45.1/32 is directly connected, Serial1/1
SP3#


SP1
SP2
SP3
SIEGE
AGENCE
SP3(config-router-af)#neighbor 1.1.1.1 activate
*Nov 4 01:23:44.511: %BGP-5-NBR_RESET: Neighbor 1.1.1.1 reset (Capability changed)
*Nov 4 01:23:44.519: %BGP-5-ADJCHANGE: neighbor 1.1.1.1 Down Capability changed
*Nov 4 01:23:44.523: %BGP_SESSION-5-ADJCHANGE: neighbor 1.1.1.1 IPv4 Unicast topology base removed from session Capability
changed
SP3(config-router-af)#neighbor 1.1.1.1 activate
*Nov 4 01:23:45.763: %BGP-5-ADJCHANGE: neighbor 1.1.1.1 Up
SP3(config-router-af)#neighbor 1.1.1.1 send-community both
SP3(config-router-af)#exit-address-family
SP3(config-router)#address-family ipv4 vrf clinet_a
SP3(config-router-af)#redistribute eigrp 100
SP3(config-router-af)#exit-address-family
SP3(config-router)#end
SP3#w
*Nov 4 01:25:00.039: %SYS-5-CONFIG_I: Configured from console by console
SP3#write memory
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
SP3#ping vrf clinet_a 192.168.45.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/94/228 ms
SP3#show ip route vrf clinet_a
Routing Table: clinet_a
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override
Gateway of last resort is not set
D 192.168.1.0/24 [90/2172416] via 192.168.45.2, 00:10:45, Serial1/1
B 192.168.12.0/24 [200/0] via 1.1.1.1, 00:08:10
192.168.45.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.45.0/24 is directly connected, Serial1/1
L 192.168.45.1/32 is directly connected, Serial1/1
SP3#ping vrf clinet_a 192.168.45.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/134/260 ms
SP3#


SP1
SP2
SP3
SIEGE
AGENCE
SP1#
*Nov 4 01:23:44.667: %BGP-5-NBR_RESET: Neighbor 3.3.3.3 reset (Peer closed the session)
*Nov 4 01:23:44.667: %BGP-5-ADJCHANGE: neighbor 3.3.3.3 Down Peer closed the session
*Nov 4 01:23:44.671: %BGP_SESSION-5-ADJCHANGE: neighbor 3.3.3.3 IPv4 Unicast topology base removed from session Peer close
d the session
SP1#
*Nov 4 01:23:45.799: %BGP-5-ADJCHANGE: neighbor 3.3.3.3 Up
SP1#ping vrf clinet_a 192.168.12.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 64/128/288 ms
SP1#show ip route vrf clinet_a

Routing Table: clinet_a
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

B   192.168.1.0/24 [200/2172416] via 3.3.3.3, 00:03:08
      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.12.0/24 is directly connected, Serial1/1
L   192.168.12.1/32 is directly connected, Serial1/1
B   192.168.45.0/24 [200/0] via 3.3.3.3, 00:03:08
SP1#show bgp vpnv4 unicast all summary
BGP router identifier 1.1.1.1, local AS number 1
BGP table version is 6, main routing table version 6
3 network entries using 468 bytes of memory
3 path entries using 240 bytes of memory
3/3 BGP path/bestpath attribute entries using 456 bytes of memory
3 BGP extended community entries using 750 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1914 total bytes of memory
BGP activity 3/0 prefixes, 3/0 paths, scan interval 60 secs

Neighbor   V   AS MsgRcvd MsgSent   TblVer InQ OutQ Up/Down State/PfxRcd
3.3.3.3   4   1   11   10   6   0   0 00:04:38   2
SP1#ping vrf clinet_a 192.168.12.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/84/144 ms
SP1#

Questions et réponses

1. SP1 reçoit une route VPNv4 avec l'attribut Route Target 100:100. D'après votre configuration, BGP va-t-il importer la route dans la table VRF du client ?

Non. La VRF importe uniquement le RT 1:100 (route-target both 1:100).

2. Y a-t-il des routes EIGRP (D) dans la table de routage VRF ?

Oui. Sur SP3 : D 192.168.1.0/24 [90/2172416] via 192.168.45.2, Serial1/1.

3. Les routeurs SIEGE et SP1 ont-ils des routes vers les réseaux du client côté AGENCE ?

Non avant la configuration de BGP. Après BGP, SP1 a : B 192.168.1.0/24 via 3.3.3.3 et B 192.168.45.0/24 via 3.3.3.3.

4. Les routeurs SP3 et AGENCE ont-ils des routes vers les réseaux du client côté SIEGE ?

Non avant la configuration de BGP. Après BGP, SP3 a : B 192.168.12.0/24 via 1.1.1.1.

5. Sur SP3, show ip route vrf : y a-t-il des routes B vers les réseaux connectés à SIEGE ? Et sur SP1 vers ceux d'AGENCE ?

Oui. SP3 : B 192.168.12.0/24 [200/0] via 1.1.1.1. SP1 : B 192.168.1.0/24 [200/2172416] et B 192.168.45.0/24 [200/0] via 3.3.3.3.